ZAKR NV-BAND

Complete Embedded AI Safety and Inference Architecture

Overview

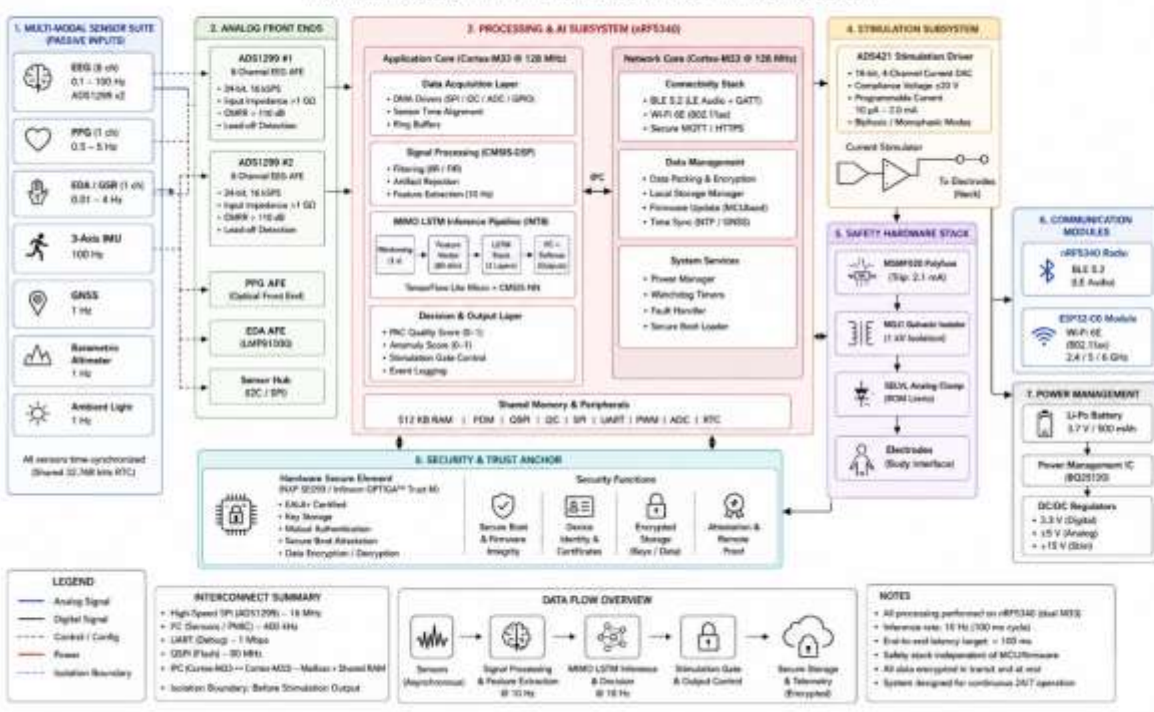
The ZAKR NV-BAND embedded architecture integrates multimodal physiological sensing, secure AI inference, ROM-based safety enforcement, and isolated neurostimulation hardware into a unified low-power wearable platform designed for continuous cognitive assistance and neuroadaptive operation.

The architecture is partitioned into six primary domains:

1. Sensor Acquisition Domain
2. Signal Conditioning and Edge AI Domain
3. Safety Enforcement Domain
4. Secure Execution Domain
5. Neurostimulation Delivery Domain
6. Audit and Provenance Domain

Figure A1. Complete Embedded System Architecture (NV-BAND)

Hardware, Processing, AI Pipeline, Safety Stack, and Communication Subsystems



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1. SENSOR ACQUISITION DOMAIN

1.1 EEG Front End

Components

- Dual ADS1299 analogue front ends
- 24-bit delta-sigma ADC
- Simultaneous multi-channel acquisition
- Differential EEG capture
- IEC 60601 isolation compliance

Signal Characteristics

- EEG channels: 8
- Sampling rate: 250 Hz
- Input noise: < 1 μVpp
- Common-mode rejection ratio: >110 dB

Electrode Topology

- T7/T8 bilateral temporal
- F3/F4 bilateral frontal
- Fz/Cz midline
- Additional IEC 10-20 validated montages

1.2 Auxiliary Sensor Suite

PPG Sensor

Purpose:

- Heart-rate variability
- Peripheral perfusion
- Sleep state estimation

Sampling:

- 100 Hz

Electrodermal Activity (EDA)

Purpose:

- Stress estimation
- Sympathetic activation
- Anxiety detection

Sampling:

- 4 Hz

GNSS Receiver

Purpose:

- Emergency localization
- Geofence monitoring
- Clinical event tagging

IMU

Components:

- 3-axis accelerometer
- 3-axis gyroscope

Purpose:

- Fall detection
- Motion artifact suppression
- Activity classification

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2. EDGE AI COMPUTE DOMAIN

2.1 Primary MCU

nRF5340 Dual Cortex-M33

Application Core:

- ARM Cortex-M33
- 128 MHz
- TensorFlow Lite Micro execution

Network Core:

- BLE communication
- Audio streaming
- Low-power coordination

Memory:

- 1 MB Flash
- 512 KB SRAM

2.2 AI Inference Stack

MIMO LSTM Architecture

Input Streams:

- EEG spectral features
- PAC metrics
- PPG variability
- EDA state
- Motion artifacts

Inference Characteristics:

- INT8 quantized execution
- CMSIS-NN optimized kernels
- 7.2 ms mean latency
- 10 Hz inference cadence

Quantization

- Post-training INT8 quantization
- Symmetric weight quantization
- Per-channel scaling
- Reduced SRAM footprint

2.3 Voice Processing Pipeline

Audio Front End

- MEMS microphone array
- ANC preprocessing
- Voice activity detection

Wakeword Detection

Standard wakeword:

- Confidence threshold: 0.85
- Sustained duration: >=1000 ms

Emergency wakeword:

- Phrase: ZAKR HELP
- Threshold: 0.72
- No sustained duration gate

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3. SAFETY ENFORCEMENT DOMAIN

3.1 Layer 1: Physical Kill Switch

Hardware

- Capacitive touch sensor
- OTP protected pattern
- Independent power path

Activation Pattern

- 3 short taps
- 2 long presses
- 3 short taps

Guardian Mode

When enabled:

- Guardian BLE proximity required
- Dual confirmation mandatory
- 60-second timeout

3.2 Layer 2: Sustained Wakeword Gate

Purpose: Prevent transient or adversarial activation.

Requirements:

- Confidence >=85%
- Duration >=1000 ms

Independent failure:

- Blocks downstream execution
- Logs rejection event

3.3 Layer 3: Prompt Integrity Verification

Mechanism

- SHA-256 hash verification
- ROM-stored reference hash
- Immutable system prompt

Validation

- Compare runtime prompt hash against ROM reference
- Any mismatch triggers halt state

3.4 Layer 4: Protocol Constraint System (PCS)

ROM-Based Template Library

15 approved protocol templates.

Validation Criteria

- Template type
- Approved montage
- Ramp-rate limits
- Session duration
- FFT waveform integrity

Rejection Conditions

- Invalid protocol structure
- Unapproved montage
- Frequency anomaly
- Constraint overflow

3.5 Layer 5: SELVL Numerical Enforcement

Absolute hard limits:

Parameter	Limit
Maximum current	2.0 mA
Maximum frequency	40 Hz
Maximum session	60 min
Sessions/day	2

All limits stored in write-protected ROM.

3.6 Layer 6: Hardware Electrical Protection

Polyfuse

- MSMF020 resettable fuse
- Trip current: ~2.1 mA

Galvanic Isolation

- MGJ1 1 kV isolation barrier

DAC

- AD5421 precision current DAC

Independent hardware cutoff possible without firmware participation.

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4. SECURE EXECUTION DOMAIN

4.1 Secure Element

Supported Devices:

- NXP SE050
- Infineon OPTIGA Trust M

Certification:

- Common Criteria EAL6+

Capabilities:

- ECDSA-P256
- Secure key storage
- Device attestation
- Cryptographic signing

4.2 Physician Authorization Flow

Clinical Protocol Requirement

Triggered when: physician_auth_required = true

Validation Steps

1. Certificate verification
2. Revocation check
3. Signature validation
4. PCS validation
5. SELVL validation
6. Execution release

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5. NEUROSTIMULATION DELIVERY DOMAIN

Signal Flow

Validated Protocol -> AD5421 DAC -> Current limiting -> Polyfuse -> Galvanic isolator -> Electrode output

Output Characteristics

- Controlled-current stimulation
- Isolated delivery path
- Hardware trip redundancy

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6. AUDIT AND PROVENANCE DOMAIN

Per-Event Audit Schema

Core Fields

- event_id
- source_type
- wakeword_ts
- protocol_json_hash
- pcs_result
- selvl_result
- device_signature

Cryptographic Integrity

- SHA-256 hashing
- ECDSA-P256 signatures
- Immutable append-only storage

Storage Security

Encryption:

- AES-256-GCM

Retention:

- Minimum 2 years

Access:

- Patient full read access
- Physician consent-based access
- Regulator local export only
- Zero ZAKR Inc. access

SYSTEM OPERATING FLOW

1. Sensor acquisition
2. Feature extraction
3. AI inference
4. Protocol generation

5. PCS validation
6. SELVL validation
7. Hardware stimulation enable
8. Audit log commit
9. Safety monitoring loop

CONCLUSION

The ZAKR NV-BAND architecture implements layered software, firmware, cryptographic, and hardware protection mechanisms designed to provide deterministic safety enforcement independent of AI model behavior. All stimulation pathways are subject to immutable ROM constraints and independent electrical cutoff hardware.